

PLANRS ACADEMY

Nurturing Scientific Inquiry & Discovery

Grade: 3 General Science

Topic: Botany Lab Guide

Duration: 30 Minutes

BUBBLES OF LIFE: WATCHING PLANTS BREATHE UNDERWATER

Hey there, Future Eco-Detectives! 

Have you ever wondered if plants breathe just like us? Since they don't have noses, it can seem like a big secret! Today, we are going to dive underwater to watch a superhero plant named **Hydrilla Verticillata** make its breath completely visible using the power of **Sunlight**!

THE STORY HOOK

The Magic Tank at the Mela 

Imagine walking through a bustling local neighborhood festival market (a *Mela*) with your grandmother (*Dadi*). Among all the bright toys and hanging plants, you spot a clear glass water container glowing under a bright lamp.

"Wow, Dadi! Look closely!" cries young Ishaan. "The green weed inside is blowing tiny, round bubbles from its cut stem, almost like it's blowing bubbles underwater! What's going on?"

Dadi smiles warmly and challenges Ishaan to become an **Eco-Detective** to solve this underwater mystery!

The Leaf Kitchen: How Leaves Cook! 🔍

To figure out the secret of the bubbles, let's look inside a leaf. Think of a leaf as a busy little kitchen (a **Rasoi**) inside the plant! Just like your family cooks delicious meals, the leaf cooks up food to help the plant grow.

The Home Rasoi 🏠	The Leaf Kitchen 🌿	What it Does! ✨
Wheat Flour & Water	Moisture from Soil & Air	The raw ingredients gathered by the roots.
Gas Stove Flame	Bright Sunlight Rays	The vital energy source used to mix items.
Delicious Rotis	Plant Growth & Oxygen Gas	The successful resulting output products!

When the sun shines bright, the leaf kitchen swings into action. The best part? The kitchen sends out clear bubbles of **Oxygen gas** into the water for us to see!

Meet Our Underwater Superhero: Hydrilla! 🦸

When land trees breathe or process sunlight, their gases blend invisibly into the air. But **Hydrilla** lives completely underwater!

- **The Trap Layer:** Because Hydrilla is fully submerged, any gases it releases get trapped by the surrounding water right away.
- **Visible Indicators:** The released oxygen forms neat, round bubbles that float straight up from the cut stem, making it easy to see how fast the leaf kitchen is working!

The Experiment: The Bubble Counter Challenge! 🕒

Let's run a three-part test as Eco-Detectives. We will keep the amount of water and the plant size exactly the same. The only thing we will change is the **Light Level!** Let's count how many bubbles appear in **10 seconds**:

Condition Alpha  Bright Sunlight Placed right next to an open window 25 Bubbles! Top Speed Production ("Tup, tup, tup!")	Condition Beta  Medium Light Placed away from windows in room center 12 Bubbles! Moderate pace with noticeable pauses	Condition Gamma  Total Darkness Covered completely by a dark box 0-2 Bubbles! Production completely halted!
--	---	---

What does this tell us? More light means the leaf kitchen works much faster! Without any light energy, the plant cannot cook or produce bubbles at all. This proves that light is mandatory!

Key Takeaways for Young Scientists 🧠

- **Plants Are Actively Alive:** Even when sitting quietly, they are constantly working, breathing, and responding to their surroundings.
- **Sunlight Powers the System:** Sunlight is the main energy fuel that drives growth.
- **Plants Help Us Breathe:** The oxygen bubbles produced by plants keep humans and animals alive. We are all beautifully connected!

Test Your Detective Skills!

Review Question 1: What gas is inside the round bubbles that rise from the cut stem of the Hydrilla plant?

 **A) Oxygen Gas (Correct! This is the essential gas humans and animals need to breathe!)**

B) Carbon Dioxide

C) Water Vapor

D) Plain Muddy Air

Review Question 2: Under which environment setup does our superhero plant work the hardest and produce the fastest stream of bubbles?

A) Deep Shade

 **B) Bright Sun (Correct! Full sunlight provides maximum energy for the leaf kitchen!)**

C) Midnight Dark

Your Outdoor Neighborhood Mission!

Scientific discovery doesn't have to end in the classroom! The next time you walk past a neighborhood park tank, a village pond (*Talab*), a stepwell (*Baoli*), or an indoor home fish tank on a sunny afternoon:

- Look closely at the submerged green weeds resting under the clear water.
- Watch for tiny, round silver pockets floating upward toward the open air.
- Observe if the bubbles slow down when a cloud temporarily blocks out the sun!
- Grab a notebook and sketch a quick picture of your findings using arrows for the bubbles!